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MILITARY SPECIFICATION

PUNCHING MACHINES, METAL, SINGLE END TYPE NUMERICALLY CONTROLLED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers numerically controlled, metal punching machines of the single end type

1.2 Classification. The machines covered by this specification shall be of the following types and sizes. The type and size machine to be supplied shall be as specified (see 6.2.1). The type shall be as listed below and the size shall be selected from table I.

TYPE I - Single punch
TYPE II - Multi-punch (Tool changer)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.1.2 Other Government documents, drawing and publications. The following other Government documents form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA - Safety and Health Standards (29 CFR 1910), General Industry

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)
ASTM-E-380-79 - Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race St. Philadelphis, PA 19103)

AMERICAN NATIONAL STANDARD & INSTITUE (ANSI)

ANSI/NFPA-79 - Electrical Standard for Metalworking Machine Tools

(Application for copies should be addressed to the American National Standard Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA) STANDARDS

RS-227	One Inch Perforated Paper Tape.
RS-244	Character Codes for Numerical Machine Tool Control Perforated Tape.

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RS-267	Axis and Motion Nomenclature for Numerically Controlled Machine Tools.
RS-274	Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/ Positioning Numerically Controlled Machines.
RS-281	Construction Standards-Numerical Machine Tool Control.
RS-358	Subset of American National Standard Code for Information Interchanged for Numerical Machine Control Perforated Tape
RS-447	Operational Command and Data Format for Numerically Controlled Machines.

(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 1-20006.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete punching machine for first article inspection and approval (see 4.3 and 6.3).

3.2 Design. The punching machine shall be new and of the manufacturers' current single end type, numerically controlled designs which meet the requirements of this specification. The machine shall be an electrically powered, motor driven unit with a solid state soft-wired, integrated circuit numerical control system. The machine's design shall include a frame with a workpiece opening or throat. The workpiece opening shall be provided with tool holders for securing the punches and dies in the punching position under a machine drive ram. Type II machine shall be designed to automatically allow the tool changing mechanism to select and rotate a punch, die and stripper assembly to a hole punching position under the machine drive ram. When specified (see 6.2.1) the machine shall be furnished with either a plasma arc or CO2 laser cutting unit. When only one type unit is acceptable, the particular cutting unit shall be as specified (see 6.2.1). The cutting unit requirements shall be fully described as

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required by paragraph 3.8. The plasma arc or CO2 laser cutting head shall be mounted between the overhang and base of the frame to form a workpiece opening. Tool changer rotation, workpiece positioning, punching action and when applicable, tool cutting head positioning and cutting shall be controlled by the numerical control system. The design shall provide convenient efficient and safe operation. The machine shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ASTM-E-380-79.

3.2.1.1 Measuring and indicating device calibrations. When specified (see 6.2.1), measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and the SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.2 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station.

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3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of "OSHA Safety and Health Standards (29 CFR 1910), General Industry" that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

3.2.5 Lubrication. An automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means for determining oil level in each reservoir shall be by oil level sight gage or by a low oil level warning light or similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

3.2.6 Interchangeability. To provide for replacements of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects such as scale and mismatching. Processes such as welding, peening, plugging or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and similar fasteners which affect machine operation or prevent loss of fluids shall be installed with means of preventing loss of tightness. Fasteners that are subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified

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herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter detrimental to painting, and at least one coat of primer and one coat of finish color is applied.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Instruction marking and indicating plates. All words on instruction and on indicating plates shall be in the English language. Characters shall be engraved, etched, embossed or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item on or to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All gears shall be machined and heat treated by a process that will impart the hardness, smoothness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear. All gears shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 COMPONENTS

3.4.1 Frame. The frame shall be a stress relieved construction of either iron or steel. Unless otherwise specified, the frame shall be of the "C" or "J" style, having a base, column, and overhang to form a workpiece opening or throat. When only one style frame is acceptable, the particular style required shall be as specified (see 6.2.1). The frame shall be sufficiently rigid to withstand stresses imposed during performance of the punching requirements specified herein. The base of the frame shall have provisions for leveling and securing the machine to a flat mounted surface. When specified (see 6.2.1), the top of the frame shall be provided with a rigging eye or other means to facilitate lifting the machine.

3.4.2 Table. The machine shall be provided with a table for supporting the workpiece in the punching operation. The table shall be constructed of iron or steel with the top surface either hardened or provided with antifriction bearings or rollers to facilitate workpiece travel across the

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table by the workpiece positioning unit. Each table motion shall operate on anti-friction bearings. The table motion and positioning shall be obtained by using anti-backlash ball screw and nuts or rack and pinion drive. Table drive shall be by servo drives and direct current (DC) motors of the totally enclosed permanent magnet type. The servo motors shall have sufficient power and torque to position the table at the speed required in table I. The table shall be of the size required for fully supporting any workpiece size of the capacity of the machine while located at any position within the range of the workpiece positioning unit. When required, the table top dimensions shall be of the size specified (see 6.2.1).

3.4.3 Machine drive. Unless otherwise specified, the punching action of the machine shall be accomplished by either a mechanical, a hydraulic, or a combination mechanical and hydraulic type drive. When required, the type drive shall be as specified (see 6.2.1).

3.4.3.1 Mechanical type drive. If a mechanical type drive is supplied, it shall be provided with an electric motor driven assembly of mechanical components with connecting shafts, gears, chains, belts or pulleys as required for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The components shall include a flywheel for supplying supplementary power to the drive motor. The flywheel shall be of cast iron or steel, and shall be dynamically balanced after machining. An eccentric or Pitman shall be provided for converting the rotary motion of the motor and flywheel to the reciprocating motion required by the ram. All rotating and sliding components shall be provided with bearings or bushings with means for positive and continuous lubrication. The energy of the drive shall be controlled by a clutch and brake arrangement that is capable of transmitting the maximum developed torque to the ram. Provisions shall be provided to automatically control the ram stroke. The ability to control the cycle by reversing under pressure shall also be provided.

3.4.3.2 Hydraulic type drive. If a hydraulic type drive is supplied, it shall be provided with a hydraulic power unit for developing and controlling the energy of a reciprocating drive ram for accomplishing punching performance specified herein. The hydraulic power unit shall include an electric motor, hydraulic pump, fluid reservoir, ram cylinder unit, and connecting hose or tubing. The fluid reservoir shall be provided with a fluid level gage, and a hydraulic fluid filter with replaceable type element. The hydraulic cylinder unit shall be the double acting type. The drive shall be provided with hydraulic fluid control devices to include a high pressure relief valve. Connecting hose(s) or tubing shall be capable of withstanding fluid pressure up to 150 percent of the rated working pressure of the power unit.

3.4.4 Tool holders. The machine shall be provided with quick change type tool holders for securing the punches and dies in the punching position under the machine ram. The tool holders shall accommodate punches and dies for punching round, square, and rectangular shaped holes of any size

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ranging up to and including the hole punching capacity for the type and size machine specified. Means shall be provided for radially locating and locking the punches and dies in the punching position. When the punches and dies are locked in the punching position, they shall be concentrically aligned within .002 inch.

3.4.5 Strippers. Machines that are not provided with self-stripping type punches shall be provided with strippers that prevent the workpiece from riding up with the punch when it is withdrawn from the die.

3.4.6 Punches and dies. When required, punches and dies of the sizes and shapes specified (see 6.2.1), shall be furnished. The punches and dies shall be tool steel, hardened and ground for accuracy. The punches and dies shall be permanently marked for size and identification.

3.4.7 Tool changer. Type II multi-punch machine shall be provided with an automatic tool changer. The machine shall be capable of automatically selecting and changing not less than 14 punch, die, and strip assemblies into the position of the punching station. The selection shall be by numerical control and manual control. Each tool assembly shall consist of a punch, stripper and die. Each tool assembly shall be capable of accepting all size tools within the capacity of the machine. Proper clamping of the tool assembly shall be provided. The machine shall automatically stop whenever a tool assembly is improperly clamped. The tool changer shall be capable of changing a tool assembly in 12 seconds or less. The tool changer may be located as a separate unit, or mounted as an integral part of the machine.

3.4.8 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to the requirements, of ANSI/NFPA No. 79. It shall be possible to operate all electrical equipment on the punching machine after connection to a single power source. The machine shall operate on both 230 and 460 volts and shall be initially wired for operation on 460 volts. All electrical connections within the equipment shall be complete and shall be made by terminals on the components, circuit boards and busses. Splicing of wiring between terminations is not permitted. All wiring, busses, terminals, circuits and units shall be clearly identifiable from the machines electrical wiring diagram.

3.4.8.1 Solid state components. Solid state design shall be used throughout for electronic components. The use of selenium or other similar rectifier aging devices is permitted only in application of voltage surge protection to other solid state components.

3.4.8.2 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearing of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure.

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3.4.8.3 Electric power disconnect. A fusible safety disconnect switch or circuit breaker rated for maximum voltage and amperage of the machine shall be provided.

3.4.8.4 Control circuit voltage. Main and auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage not exceeding 115 volts.

3.4.9 Numerical control system. The numerical control shall be of the solid state, soft-wired, integrated circuit, closed loop type. The system shall provide automatic control of machine functions, operating modes, punch selection and two axes control of the horizontal "X" and "Y" motion. When a plasma - arc or CO2 laser cutting head is furnished, the cutting head shall also be controlled by the numerical control unit. The axes shall be identified in accordance with EIA standard RS-267. The controls shall be a positioning and controuring type having linear and circular interpolation and be capable of simultaneous two axis controuring. Control features shall include buffer storage, part program storage, program edit capability and control diagnostics. The control unit shall direct machine functions from part program data in memory storage, punched tape, and manual data input. The control shall initiate a halt or an error signal should a fault condition occur in the control. All necessary executive program routines shall be furnished in the form of 1 inch, eight channel, binary coded decimal punched tape for controls having volatile memory. The control unit shall conform to EIA Standard RS-281. A line voltage variation of $\pm$ 10% from normal shall not adversely affect the control system functions. The control shall be capable of functioning in ambient temperatures ranging from 50°F to 120°F and relative humidities ranging from 5% to 95%. The control shall automatically shut down when internal operating temperatures exceed safe operating temperature. Control resolution shall be not greater than .001 inch for each linear axis. The auxiliary functions of the system shall include, but not be limited to punch action, and opening and closing of the workpiece clamps. The system shall be provided with operational interlocks and functional protection to prevent the machine from operating when the punch assembly is not locked in the punching position, the machine drive brake is not released or the air or hydraulic pressure of the machine or auxiliary drive units drops below the required operating level. Interlock circuitry shall also be provided for assuring sequential operation of the system. The control unit shall be provided with a two-axis contouring and point-to-point positioning capabilites and with absolute and incremental programming. The control unit shall rotate the tool changer, if applicable, lock the programmed punch and die station in the punching position, and position the workpiece throughout the full range of travel along the "X" and "Y" axes of motion. The control unit shall automatically control the acceleration, and deceleration of all axes of motion. The control unit shall control axis movements individually, simultaneously, or in any combination. The control unit shall control the positioning units for accomplishing any function within the capacity of the machine without loss of cycle synchronization.

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The smallest program increment for positioning the workpiece shall be not more than 0.001 inch and the largest shall be not less than the workpiece positioning range specified in table I.

3.4.9.1 Data input. The data input media shall be one inch, eight channel, punched tape in conformance with EIA Standards RS-227, RS-244, RS-274, RS-358 and RS-447.

3.4.9.1.1 Tape reader. A photoelectric tape reader shall be provided. The tape reader shall be capable of holding, feeding and reading tape prepared in accordance with paragraph 3.4.9.1, at a rate of not less than 150 characters per second. The tape reader shall be supplied with standard feed and take-up spoolers that are not less than 5-1/4 inches in diameter. The tape reader shall be housed in a dust-free area of the machine control unit and shall be easily accessible for maintenance and tape handling. The control shall have a tape error detection capability to continuously check for the following errors:

a. Odd parity per EIA RS-244
b. Even parity per EIA RS-358
c. Misaligned sprocket holes
d. No sprocket holes

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display unit to indicate fault condition.

3.4.9.1.2 Manual data input. A manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

3.4.9.2 Control Panel. The control unit shall have an operator control panel for displaying the control functions and visual signals of the numerical control system and the machine tool. The control panel shall be provided with knobs, switches or digital input for controlling system functions and shall include at least power-on and off, cycle start-stop, emergency stop, optional stop, mode selection, feed hold, feedrate override, jog, jog direction, and end of block. Indicator lights or the data display unit shall indicate or display at least reading error, in-position, over--temperature, tool change if applicable, program stop, end of program, punch assembly not locked in punching position, and low air or hydraulic pressure.

3.4.9.3 Data display. Unless otherwise specified, either a cathode ray tube (CRT) or a universal alphanumeric data display unit shall be provided. When only one type of data display is acceptable, the particular type shall be specified (see 6.2.1). The CRT unit shall be capable of displaying at least 250 characters simultaneously. The universal alphanumeric display

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unit shall be capable of being used in a variable scanning mode to interrogate all registers sequentially or held to display only a particular register as the content changes. Either unit shall display, as a minimum, the following data:

a. Sequence number readout.
b. Actual position of each axis. The readout shall at least display the accumulative command axis position of one axis at a time without loss of reference dimensions as a result of the switching.
c. Control and system error message.
d. Miscellaneous functions.

3.4.9.4 Control system modes of operation. The operational modes of the control unit shall include manual, automatic, single block, manual data input, and manual jog.

3.4.9.5 Absolute/incremental input. The control shall have absolute/-incremental input capability. This mode shall be selected by a preparatory "G" code which may be switched during the tape program.

3.4.9.6 Automatic tape code recognition. The control shall have the capability to automatically recognize and read tape punched with either EIA RS-244 or RS-358 (ASCII) character codes.

3.4.9.7 Switchable inch/metric input. The control system shall provide switchable inch/metric input capability.

3.4.9.8 Part program storage. Unless otherwise specified (see 6.2.1), the machine control unit shall be capable of storing in memory the content of not less than 150 feet of postprocessed part program tape data.

3.4.9.9 Standby battery power. Standby battery power shall be provided for controls having volatile memory to hold the size memory provided at full load for a period of not less than 72 hours when the primary power source fails.

3.4.9.10 Part program edit. A program edit capability shall be provided to allow block insertion, deletion or modification of any part program in the part program memory. Data related to part program edit shall be input by the manual data input provided on the control panel. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.4.9.11 Contouring Feed rate override. A feed rate override control shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that programed.

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3.4.9.12 Sequence number search. The control shall have the capability to permit the operator to search for any desired sequence number.

3.4.9.13 Mirror image. The machine shall be capable of producing either right or left-hand parts from the same tape data through the use of inversion switches for each axis. Inversion of each axis may be by keyboard entry.

3.4.9.14 Preparatory function (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.9.15 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274.

3.4.9.16 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.4.9.17 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall delay or store miltiple blocks of data until required by the controlling devices.

3.4.9.18 Block-by-block read. The control shall have a block-by-block read function to provide for reading information one block at a time.

3.4.9.19 Lead and reversal error compensation. Machines equipped with ball screw and nut assembly shall have lead error and reversal error compensation to permit data to be stored in the control to compensate for lead screw errors and backlash of each linear axis.

3.4.9.20 Additional features. The following additional numerical control features shall be provided:

a. Four quadrant programming.
b. Incremental and continuous jog for each axis.
c. Leading zero suppression or decimal point programming.
d. Inches per minute (IPM)/Millimeters Per Minute (MMPM), programming in all modes of operation.
e. Pattern repeat.
f. Tape edit keylock.

3.4.9.21 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.4.9.22 Set zero. The control unit shall have a "set zero" feature to permit the specified axis register to be reset to zero establishing present axis location as zero.

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3.4.9.23 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to effectively operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the controls, machine tool, part program and operator's actions.

3.4.9.24 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished to provide for diagnostic checking of the machine control unit. The diagnostic software shall test, exercise and display failures at least to board level.

3.4.9.25 Positioning units. The primary command functions of the system shall be accomplished by workpiece postioning units with positioning range and speed to comply with table I. Positioning unit travel shall be obtained by electric motor driven anti-backlash ball screws and nuts or rack and pinion drive. Response of the positioning units to the primary commands for starting and stopping and the rate of motion shall be such that the specified punching accuracy is maintained. The workpiece positioning unit shall hold the workpiece by at least two clamps, which shall be adjustable for holding any workpiece size ranging up to the capacity of the machine. The workpiece positioning unit shall be provided with means to prevent the workpiece clamps from being punched during the punching operation. When specified (see 6.2.1), the workpiece positioning unit shall be provided with automatic sheet repositioner characteristics for location of sheets longer than the rated capacity of table I by tape command. The closed loop workpiece positioning unit shall measure and feed back to the control unit, output signals of actual workpiece position that shall be compared with the input positioning commands of the program tape data. The control unit shall be arranged to shut down the machine for positioning errors in excess of 0.003 inch.

3.4.9.26 Postprocessor. When specified (see 6.2.1) a postprocessor shall be furnished with the capability of converting the coordinate data generated by the computer program into machine language format of the punching machine. The computer and operating softwire, on which the postprocessor is to operate shall be fully described under 6.2.1(u).

3.4.9.27 Test tape. Unless otherwise specified (see 6.2.1), a test tape and printout programmed to perform all system functions specified herein shall be furnished. The test tape shall operate all slides, motors, pumps adjusting mechanisms, and accessories during each cycle of the test tape.

3.5 Sizes and characteristics. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall not be less than those specified in table I or as applicable for the machine size specified.

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TABLE I - SIZES AND CHARACTERISTICS

Characteristics	Size 1	Size 2	Size 3	Size 4	Size 5	Size 6	Size 7	Size 8	Size 9
Throat depth inches	30	30	48	60	60	48	60	50	50
Workpiece positioning unit range (without repositioning the workpiece within the workpiece positioning unit):									
In "X" axis, inches	30	50	60	70	80	60	80	85	80
In "Y" axis, inches	30	30	48	60	60	48	60	50	50
Workpiece positioning speed along the "X" and "Y" axis, inches per minute	400	400	400	400	400	600	500	600	500
Hole punching capacity in SAE 1020 steel:									
maximum diameter, inches	3.000	3.000	3.000	4.000	4.000	4.000	4.000	5.000	6.000
in maximum thickness inches	.135	.135	.060	.120	.100	.135	.250	.135	.375
Diameter, inches	1.000	1.000	1.000	.500	1.000	.500	1.000	1.500	1.625
in maximum material thickness, inches	.250	.250	.250	.375	.375	.500	.500	.500	1.125
Holes per minute spaced on one inch (two inch for size 9) centers in the maximum material thickness	100	100	150	100	75	50	75	55	35
Punched hole centerline dimension accuracy inches	±.005	±.005	±.005	±.005	±.005	±.005	±.005	±.005	±.005
Repeatability of hole location	±.003	±.003	±.003	±.003	±.003	±.003	±.003	±.003	±.003
Workpiece size, inches	30x30	30x48	48x60	60x70	60x80	48x60	60x80	50x80	50x80

Requirements are not less than the figures shown, except ± items shall be as stated.

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3.6 Performance. The machine shall be capable of automatic punching of holes in SAE 1020 steel to meet the capacity and accuracy requirements of table I as applicable for the machine size specified. The numerical control system shall meet the performance requirements of 3.4.9.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), one set of hand tools, normally furnished by the manufcturer, shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.10 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

3.11 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided that the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

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Punching capacity, maximum hole diameter

Punching capacity, maximum material thickness

Power input characteristics and rating

Date of Manufacture

Contract Number or Order Number

National Stock Number or Plant Equipment Code

3.12 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.13 Workmanship. Workmanship of the machines and accessories shall be commensurate with the requirements specified herein, and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4.).

4.3 First article inspection. First article inspection shall be applied to the first article when required in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests in 4.6. Failure of the first article to pass the examination or any first article tests shall be cause for rejection.

4.4. Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in

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4.6.1, and the inspection in 4.6.5. Failure of the machine to pass the examination, test, or inspection shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.7 through 3.13.

4.6 Tests.

4.6.1 Operational test. The machine shall be operated at no load for not less than one hour, during which time the machine drive, and if applicable the tool changer shall be actuated and observed to determine compliance with the rquirements specified hrein. The numerical control system shall be operated in accordance with the manufcturer's standard operating test procedure for warm-up and run-off checks. Run-off checks of all mechanical functions of the machine and numerical control system functions shall be made in each mode of operation specified herein. Upon completion of the warm-up check procedure, the system shall be cycled continuously for a period of not less than two hours under tape control while programmed to demonstrate reliability of the equipment. There shall be no failure or malfunction of the machine or control. Should a failure or malfunction occur, the test shall be restarted from setpoint and retimed from zero hours. The program tape used for cycling shall position the machine over the entire working range and shall include operation of all controlled functions of the machine. The restart of each cycle shall occur within one minute of completion. At the end of the two hour test period, the machine shall return to setpoint within as accuracy of plus or minus 0.005 inch. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.6.1.1 Punch and die station alignment. Each punch and corresponding die station shall be positioned and locked in the punching position, and then checked to verify the punch and die alignment requirements of 3.4.4.

4.6.2 Machine performance test. The machine shall be tested as follows to determine compliance with the performance rquirements of table I as applicable for the size machine specified. When hole punching is required, SAE 1020 steel shall be used.

4.6.2.1 Positioning speed. Positioning speed shall be verified by clamping the maximum capacity size workpiece in the workpiece through the full positioning range for the "X" and "Y" axes. The time required and distance traveled shall be compared to determine compliance with the specified requirement.

4.6.2.2 Hole punching capacity. Hole punching capacity shall be verified by operating the machine for punching at least 12 holes of each of the diameters specified in table I for each of the material thicknesses specified.

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4.6.2.3 Hole punching speed and accuracy. Hole punching speed and accuracy shall be verified by oprating the machine to continuously punch 0.500 inch diamter holes, spaced on one inch centers, in a workpiece with a thickness of not less than the maximum material thickness capacity of the applicable size machine. The holes shall be punched for a period of one minute. Upon completion, the number of holes punched shall be counted and the resulting punching accuracy shall be measured to determine compliance with the specified requirements.

4.6.2.4 Repeatability test. Two workpieces of 1/8" thick material shall be punched in consecutive order with 24 holes each utilizing a program tape with a hole pattern of the suppliers choice. The two plates shall be inspected to verify the repeatability requirements of table I.

4.6.3 Numerical control performance test. The numerical control system shall be tested by actuating the control unit to operate the machine under the automatic, semi-autmatic, and manual modes.

4.6.3.1 Automatic. The test tape specified in 3.4.9.28 shall be inserted in the control unit and the numerical control system operated to perform the functions specified. The printout furnished shall be used to verify the specified functions.

4.6.3.2 Semi-automatic. The control unit shall be operated as specified in 4.6.3.1, but with at least six separate insertions of commands that require the machine to perform positioning functions. Each function shall be verified for accuracy.

4.6.3.3 Manual. The control unit shall be manually operated to verify each of the primary and auxiliary functions of the system.

4.6.4 Postprocessor validation. When a postprocessor is furnished (see 3.4.9.26), the supplier is responsible for having the postprocessor validated on the user's computer to insure compatibility with the hardware and software approximately 30 days prior to deliver to the NC machine. One or more test tapes shall be processed form the manuscript provided by the supplier to exercise all programmable functions provided in the control and to produce a test part which demonstrates all machining capabilities.

4.6.5 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall conform to the requirements of MIL-STD-1188.

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6. NOTES

6.1 Intended use. The machines covered by this specification are for use in producing complex punched configurations in sheet metal to close tolerances by automatic control of the machine. The machine can be used to produce prototype parts from a taped program, or to produce items repetitively from a proven process to established accuracies.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2 and table 1).

c. Specification and standards, if different (see 2.1.1).

d. When first article is required for inspection and approval. (See 3.1 and 4.3).

e. Cutting unit, if required and if a specific type, state required type (see 3.2).

f. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

g. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (see 3.2.1.1).

h. Dual calibrations (US and SI); if required (see 3.2.1.2).

i. Specify additional safety and health requirements (see 3.2.4).

j. Painting, if different (see 3.3.5).

k. If machine frame style is to be a specific type, state required style (see 3.4.1).

l. Rigging eye, if required (see 3.4.1).

m. Table top dimensions, if different (see 3.4.2).

n. If machine drive is to be a specific type, state required type (see 3.4.3).

o. Punches and dies, specify shapes and sizes required (see 3.4.6).

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p. Electrical equipment and input power supply, if different (see 3.4.8).

q. Motor enclosure, if different (see 3.4.8.2).

r. Data display, if to be of a specific type, state required type (see 3.4.9.3).

s. Part program storage, if different (see 3.4.9.8).

t. Automatic sheet repositioner, if required (see 3.4.9.26)

u. Postprocessor, if required, specify computer name, model and software required (see 3.4.9.26).

v. Test tape, if different (see 3.4.9.28).

w. Size and characteristics, if different (see 3.5.).

x. Standard equipment, if different (see 3.7).

y. Optional equipment, if required (see 3.8).

z. List of repair parts required (see 3.9).

aa. Lubrication chart, if different (see 3.10).

bb. Nameplate, if different (see 3.11).

cc. First Article inspection, if different (see 4.3.).

dd. Quality conformance inspection, if different (see 4.4).

ee. Level of packaging and marking required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as opertor's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664 should be specified on a DD Form 1423 incorporated into the contract.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquistion Regulations. The first article should be a reproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation

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levels, electromagnetic emissions, noxious vaports, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring agency is advised to analyze the existing hazard level in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of-operation guarding is required, the procuring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(i).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.6 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-P-80172A	MIL-P-80172B
Type I-Closed Loop N/C (deleted)	Type I-Single punch (New type)
Type II-Closed Loop or Open Loop N/C (deleted)	Type II-Milti-punch (New type)
Size 1 (Deleted)	
Size 2	Size 1
Size 3	Size 2
	Size 3 (New size)
	Size 4 (New size)
	Size 5 (New size)
Size 4	Size 6
	Size 7 (New size)
Size 5	Size 8
Size 6	Size 9
Size 4	Size 6
	Size 7 (New size)
Size 5	Size 8
Size 6	Size 9

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review Activities:
DLA-GS
Air Force - 84
User Activities:
Navy - MC

Preparing Activity:
DLA - IP

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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